

Case 1 { when $U \cap V \neq \emptyset$ }

Suppose U and V are subspaces of a finite dimensional vector space with $\dim(U) = r$ and $\dim(V) = s$

Assume that U and V are finite-dimensional vector spaces

Let $B_U = \{B_1, B_2, \dots, B_r\}$ be the basis in U and $B_V = \{\alpha_1, \alpha_2, \dots, \alpha_s\}$ be the basis in V . Then for each vector in B_V , if it is linearly expressible in terms of the elements of B_U , then we replace any one vector in B_U with the former vector from B_V . When the algorithm terminates and if B_V and B_U contain p elements in common with each other, then we have that $\dim(U \cap V) = p$

$\Rightarrow U \cup V$ contains $r + s - p$ elements

Thus $\dim(U + V) = \dim(U) + \dim(V) - \dim(U \cap V)$

$$\therefore \dim(U + V) = r + s - p.$$

Case 2 { when $U \cap V = \emptyset$ }

Suppose U and V are subspaces of a finite dimensional vector space with $\dim(U) = r$ and $\dim(V) = s$

as $U \cap V = \emptyset$, $\Rightarrow U$ and V are direct sum

i.e. U and V are independent subspaces and are also

complementary subspaces, so the basis for $U \oplus V$ is the set $\{B_1, B_2, \dots, B_r, \alpha_1, \alpha_2, \dots, \alpha_s\}$

$$\begin{aligned} \Rightarrow \dim(U \oplus V) &= \dim(U) + \dim(V) \\ &= r + s. \end{aligned}$$

B1 (b)

We have that

$$U = \text{span}(u_1, u_2, u_3)$$

$$V = \text{span}(v_1, v_2, v_3)$$

Finding $\dim(U)$. Consider the matrix $[u_1 \ u_2 \ u_3]$.

Take $A = [u_1 \ u_2 \ u_3] = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

RRFF(A) follows as

$$R_2 \rightarrow R_2 - R_1 \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow R_3 \rightarrow R_3 - R_2 \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow R_4 \rightarrow R_4 - R_3 \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Since u_1, u_2, u_3 are independent it follows that $\dim(U) = 3$

Finding $\dim(V)$. Consider the $[v_1 \ v_2 \ v_3]$. Let $B = [v_1 \ v_2 \ v_3]$

$$B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 2 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

RRFF(B) follows that

$$R_3 \rightarrow R_3 - R_1 \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 2 \\ 0 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix} \rightarrow \frac{1}{2} R_2 \rightarrow R_2 \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{matrix} R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2 \end{matrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow$$

$\rightarrow R_4 \rightarrow R_4 - R_3 \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Since v_1, v_2, v_3 are independent then it follows that $\dim(V) = 3$.

Then it follows that

$$\begin{aligned}\dim(U+V) &= \dim(U) + \dim(V) - \dim(U \cap V) \\ &= 3 + 3 - 2 \\ &= 4\end{aligned}$$

where $\alpha = \rho(x, y) |x, y > 0, y \in \mathbb{R}$ where

$(x_1, y_1) + (x_2, y_2) = (x_1 x_2, y_1 y_2)$ for vector addition

$t(x_1, y_1) = (x_1^t, y_1^t)$ for scalar multiplication

Let $u, v, w \in Q$ such that $u = (x_1, y_1)$, $v = (x_2, y_2)$ and $w = (x_3, y_3)$

(a) (i) Testing for closure under vector addition

$$u + v = (x_1 x_2, y_1 y_2) \in Q \quad x_i, y_i > 0 \text{ for } i=1, 2$$

(ii) Testing for commutativity

$$\begin{aligned} u + v &= (x_1 x_2, y_1 y_2) \\ &= (x_2 x_1, y_2 y_1) \\ &= v + u \end{aligned}$$

Alternatively

$$\begin{aligned} u + v &= (x_1 x_2, y_1 y_2) \\ v + u &= (x_2 x_1, y_2 y_1) \\ \Rightarrow u + v &= v + u. \end{aligned}$$

(iii) Testing for associativity

$$\begin{aligned} (u + v) + w &= (x_1 x_2, y_1 y_2) + (x_3, y_3) \\ &= (x_1 x_2 x_3, y_1 y_2 y_3) \\ &= (x_1, y_1) + (x_2 x_3, y_2 y_3) \\ &= u + (v + w) \end{aligned}$$

Alt

$$\begin{aligned} (u + v) + w &= (x_1 x_2, y_1 y_2) + (x_3, y_3) \\ &= (x_1 x_2 x_3, y_1 y_2 y_3) \end{aligned}$$

$$\begin{aligned} u + (v + w) &= (x_1, y_1) + (x_2 x_3, y_2 y_3) \\ &= (x_1 x_2 x_3, y_1 y_2 y_3) \end{aligned}$$

$$\Rightarrow (u + v) + w = u + (v + w)$$

(iv) Testing for additive identity

Suppose that the additive identity element is $(1, 1)$.

Then it follows that

$$\begin{aligned}u + (1, 1) &= (x_1, y_1) + (1, 1) \\&= (x_1, y_1) \\&= u\end{aligned}$$

Therefore $(1, 1)$ is the additive identity.

(v) For the additive inverse,

We have

$$\begin{aligned}u + (-u) &= (x_1, y_1) + (-x_1^{-1}, -y_1^{-1}) \\&= (x_1, y_1) + (x_1^{-1}, y_1^{-1}) \\&= (1, 1)\end{aligned}$$

After

$$u + (-u) = (1, 1)$$

$$(x_1, y_1) + (u_1, u_2) = (1, 1)$$

$$(x_1 u_1, y_1 u_2) = (1, 1)$$

$$\begin{aligned}\Rightarrow (u_1, u_2) &= \left(\frac{1}{x_1}, \frac{1}{y_1}\right) \\&= (x_1^{-1}, y_1^{-1})\end{aligned}$$

$$\therefore \forall (x_1, y_1) \in Q \quad \exists (x_1^{-1}, y_1^{-1}) \text{ s.t. } (x_1, y_1) + (x_1^{-1}, y_1^{-1}) = (1, 1)$$

(vi) Testing for closure under scalar multiplication.

$$\begin{aligned}\lambda(v) &= \lambda(x_1, y_1) \\&= (x_1^\lambda, y_1^\lambda) \in Q \text{ since } x_1^\lambda, y_1^\lambda > 0.\end{aligned}$$

(vii) Let $\lambda, \mu \in \mathbb{R}$. Then

$$\begin{aligned}(\lambda + \mu)v &= (\lambda + \mu)(x_1, y_1) = (x_1^{(\lambda + \mu)}, y_1^{(\lambda + \mu)}) \\&= (x_1^\lambda \cdot x_1^\mu, y_1^\lambda \cdot y_1^\mu) \\&= (x_1^\lambda, y_1^\lambda) + (x_1^\mu, y_1^\mu) = \lambda v + \mu v\end{aligned}$$

$$\begin{aligned}
 (ii) \quad \lambda(v+u) &= \lambda(x_1, x_2, y_1, y_2) \\
 &= ((x_1, x_2)^\lambda, (y_1, y_2)^\lambda) \\
 &= (x_1^\lambda, x_2^\lambda, y_1^\lambda, y_2^\lambda) \\
 &= (x_1^\lambda, y_1^\lambda) + (x_2^\lambda, y_2^\lambda) \\
 &= \lambda(x_1, y_1) + \lambda(x_2, y_2) \\
 &= \lambda v + \lambda u
 \end{aligned}$$

$$\begin{aligned}
 (ix) \quad (\lambda v)^r &= [(x_1)^{\lambda r}, (y_1)^{\lambda r}] \\
 &= [((x_1)^\lambda)^r, ((y_1)^\lambda)^r] \\
 &= r(x_1^\lambda, y_1^\lambda) \\
 &= r(\lambda v)
 \end{aligned}$$

$$\begin{aligned}
 (x) \quad 1 \cdot v &= 1(x_1, y_1) \\
 &= (x_1, y_1) \\
 &= v \\
 &= v.
 \end{aligned}$$

(b) $\mathcal{Q}$ is a vector space over $\mathbb{R}$.

$$(c) \quad S = \{(x, y, z) : 2x + y - z = 0\}$$

let $v, u \in S$ such that

$$v = (x_1, y_1, z_1) \text{ s.t. } 2x_1 + y_1 - z_1 = 0$$

$$u = (x_2, y_2, z_2) \text{ s.t. } 2x_2 + y_2 - z_2 = 0$$

We have that $(0, 0, 0) \in S$ _{since} $2(0) + 0 - 0 = 0$

Testing for closure

$$v + u = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$$

$$\begin{aligned} \text{But } 2(x_1 + x_2) + y_1 + y_2 - (z_1 + z_2) &= 2x_1 + y_1 - z_1 + 2x_2 + y_2 - z_2 \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

$$\Rightarrow v + u \in Q$$

Let $\lambda \in \mathbb{R}$.

$$\begin{aligned} \lambda u &= \lambda(x_1, y_1, z_1) \\ &= (\lambda x_1, \lambda y_1, \lambda z_1) \end{aligned}$$

$$\begin{aligned} \text{But } 2\lambda x_1 + \lambda y_1 - \lambda z_1 &= \lambda(2x_1 + y_1 - z_1) \\ &= \lambda(0) \\ &= 0 \end{aligned}$$

$\Rightarrow \lambda u \in S$. Hence S is a subspace

Alternatively

$$\begin{aligned} S &= \{(x, y, z) \mid 2x + y - z = 0\} \\ &= \{(x, y, z) \mid y = z - 2x\} \end{aligned}$$

Let $u, v \in S$. Let $v = (x_1, y_1, z_1)$ and $u = (x_2, y_2, z_2)$
where $y_1 = z_1 - 2x_1$ and $y_2 = z_2 - 2x_2$

$$\text{Now } u + v = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$$

$$\begin{aligned} \text{Then } y_1 + y_2 &= z_1 - 2x_1 + z_2 - 2x_2 \\ &= (z_1 + z_2) - 2(x_1 + x_2) \end{aligned}$$

$$\Rightarrow u + v \in Q$$

$$\begin{aligned} \lambda u &= \lambda(x_2, y_2, z_2) \\ &= (\lambda x_2, \lambda y_2, \lambda z_2) \end{aligned}$$

$$\begin{aligned} \lambda y_2 &= \lambda(z_2 - 2x_2) \\ &= \lambda z_2 - 2\lambda x_2 \\ &= \lambda z_2 - 2\lambda x_2 \end{aligned}$$

$\Rightarrow \lambda u \in Q$. Hence Q is a subspace. Page 8

① $2x_1 - 8x_2 - 7x_3 + 5x_4 + 2x_5 = -2$

$x_1 - 2x_2 - 4x_3 + 3x_4 + x_5 = -2$

$2x_1 - 4x_3 + 2x_4 + x_5 = 3$

$x_1 - 5x_2 - 7x_3 + 6x_4 + 2x_5 = -7$

* Continuation

$$\Rightarrow \left[\begin{array}{ccccc|c} 2 & -3 & -7 & 5 & 2 & -2 \\ 1 & -2 & -4 & 3 & 1 & -2 \\ 2 & 0 & -4 & 2 & 1 & 3 \\ 1 & -5 & -7 & 6 & 2 & -7 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2 + R_4} \left[\begin{array}{ccccc|c} 2 & 0 & -4 & 2 & 2 & 4 \\ 0 & 1 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\begin{array}{l} R_1 \rightarrow R_1 - 2R_2 \\ R_3 \rightarrow R_3 - R_1 \\ R_4 \rightarrow R_4 - 2R_2 \end{array} \left[\begin{array}{ccccc|c} 2 & -3 & -7 & 5 & 2 & -2 \\ 0 & 1 & 1 & -1 & 0 & 2 \\ 0 & 3 & 3 & -3 & -1 & 5 \\ 0 & 7 & 7 & -7 & -2 & 12 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccccc|c} 1 & 0 & -2 & 1 & 1 & 2 \\ 0 & 1 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_4 \rightarrow R_4 - 7R_1 \end{array} \left[\begin{array}{ccccc|c} 2 & -3 & -7 & 5 & 2 & -2 \\ 0 & 1 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & -2 & -2 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccccc|c} 1 & 0 & -2 & 1 & 0 & 1 \\ 0 & 1 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Now

$x_5 = 1 ; x_2 + x_3 - x_4 = 2$

$x_1 - 2x_3 + x_4 = 1$

Let $x_3 = t_1 ; x_4 = t_2$ $t_1, t_2 \in \mathbb{R}$

Then

$x_2 = 2 - t_1 + t_2$

$x_1 = 1 + 2t_1 - t_2$

$x_3 = t_1$

$x_4 = t_2$

$x_5 = 1$

Thus the system has infinitely many solutions

$$\begin{array}{l} R_3 \rightarrow -R_3 \\ R_4 \rightarrow -\frac{1}{2}R_4 \end{array} \left[\begin{array}{ccccc|c} 2 & -3 & -7 & 5 & 2 & -2 \\ 0 & 1 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array} \right] \xrightarrow{R_4 \rightarrow R_4 - R_3} \left[\begin{array}{ccccc|c} 2 & -3 & -7 & 5 & 2 & -2 \\ 0 & 1 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] *$$

B3 (b)

It is enough to find some scalars $\lambda, \mu \in \mathbb{R}$ s.t.

$$\begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} = \lambda \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \left[\begin{array}{cc|c} 2 & 1 & 1 \\ 1 & -1 & 0 \\ -1 & 1 & 3 \end{array} \right]$$

$$\begin{array}{l} R_1 \rightarrow R_1 - 2R_2 \\ R_3 \rightarrow R_3 + 2R_2 \end{array} \left[\begin{array}{cc|c} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 3 & 7 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2 \left[\begin{array}{cc|c} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 6 \end{array} \right]$$

$$\Rightarrow \lambda(0) + \mu(0) = 6$$

$$\Rightarrow 0 = 6 \text{ which is false.}$$

Therefore we can conclude that $\begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$ is not in the span of $\left\{ \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

SKUDN

1. ~~True~~ ^{False}
 2. True
 3. False
 4. True
 5. True
 6. ~~False~~ ^{True}
 7. False
 8. True
 9. False
 10. True
 11. True
 12. True
 13. False
 14. False
 15. True
 16. True
 17. True
 18. False
 19. True
 20. False
 21. False
 22. True
 23. False
 24. False
 25. True

BH

$$C_1 \underline{x} + C_2 \underline{x} = 0$$

$$C_1 \begin{bmatrix} -2 \\ -1 \\ 3 \\ 4 \end{bmatrix} + C_2 \begin{bmatrix} -8 \\ -2 \\ -2 \\ 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} -2 & -8 \\ -1 & -2 \\ 3 & -2 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} -2 & -8 & 0 \\ -1 & -2 & 0 \\ 3 & -2 & 0 \\ 4 & 1 & 0 \end{array} \right]$$

$$R_2 - \frac{1}{2}R_1 \rightarrow R_2, R_3 + \frac{3}{2}R_1 \rightarrow R_3, R_4 + 2R_1 \rightarrow R_4$$

$$\left[\begin{array}{cc|c} -2 & -8 & 0 \\ 0 & 2 & 0 \\ 0 & -9 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

$$\begin{bmatrix} -2 & -8 & 10 & -2 \\ 0 & 2 & 1 & -1 \\ 0 & -14 & 1 & 3 \\ 0 & -15 & 6 & 4 \end{bmatrix}$$

$$-15c_2 = 0 \Rightarrow c_2 = 0$$

$$-2c_1 - 8c_2 = 0 \Rightarrow c_1 = 0$$

Hence Yes, the 2 vectors are linearly independent

$$\text{Also: } \underline{v}_1 \cdot \underline{v}_2 = 0$$

$$\begin{bmatrix} -2 \\ -1 \\ 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} -8 \\ -2 \\ -2 \\ 1 \end{bmatrix} = 16 \neq 0$$

Hence the vectors are not orthogonal (No)

$$(b) \quad c_1 \underline{w}_1 + c_2 \underline{w}_2 + c_3 \underline{w}_3 + c_4 \underline{w}_4 = \underline{0}$$

$$\begin{bmatrix} -2 & 8 & 10 & -2 \\ -1 & 2 & 1 & -1 \\ 3 & 2 & 1 & 3 \\ 4 & 1 & 6 & 4 \end{bmatrix} \begin{matrix} 0 \\ 0 \\ 0 \\ 0 \end{matrix}$$

$$R_2 - \frac{1}{2}R_1 \rightarrow R_2, \quad R_3 + \frac{3}{2}R_1 \rightarrow R_3, \quad R_4 + 2R_1 \rightarrow R_4$$

$$\begin{bmatrix} -2 & 8 & 10 & -2 \\ 0 & -2 & -4 & 0 \\ 0 & 14 & 16 & 0 \\ 0 & 17 & 26 & 0 \end{bmatrix} \begin{matrix} 0 \\ 0 \\ 0 \\ 0 \end{matrix}$$

$$R_3 + 7R_2 \rightarrow R_3, \quad R_4 + \frac{17}{2}R_2 \rightarrow R_4$$

$$\begin{bmatrix} 2 & 8 & 10 & -2 & 0 \\ 0 & -2 & -4 & 0 & 0 \\ 0 & 0 & -12 & 0 & 0 \\ 0 & 0 & -8 & 0 & 0 \end{bmatrix}$$

We have that $c_3 = 0$

$$-2c_2 - 4c_3 = 0 \Rightarrow c_2 = 0$$

$$-2c_4 + 8c_2 + 10c_3 - 2c_4 = 0$$

$$-2c_4 - 2c_4 = 0$$

$$c_4 = -c_4$$

OR

$$c_4 = -c_4 \Rightarrow$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} -c_4 \\ 0 \\ 0 \\ c_4 \end{bmatrix} \Rightarrow c_4 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Since, the ~~sets~~ c_i are not all zero, we have that the vectors are not linearly independent, so the set is not Basis.

(c) Let $B = \frac{1}{2} A$

$$B = \begin{bmatrix} 0 & 0 & 0 & 1/2 \\ 0 & 0 & 1 & 1 \\ 0 & 3/2 & 7/2 & 3/2 \\ 2 & 2 & 2 & 2 \end{bmatrix}$$

$$\det(B) = \frac{1}{2} \times 1 \times \frac{3}{7} \times 7$$

$$= \frac{3}{2}$$

$$=$$

NB: When the Question B4(b) Someone can find the determinants and make the decision based on that. If the determinant is not equal to zero, then they are independent. From the Matrix, the determinant is 0. Hence, the set is not a basis.

- | | | |
|------------------|-------------------|-----------------|
| 1 B | 11 A | 21 B |
| 2 A | 12 B A | 22 A |
| 3 B | 13 B | 23 B |
| 4 A | 14 B | 24 B |
| 5 A | 15 A | 25 A |
| 6 A B | 16 A | |
| 7 B | 17 A | |
| 8 B A | 18 B | |
| 9 B | 19 A | |
| 10 A | 20 A B | |